

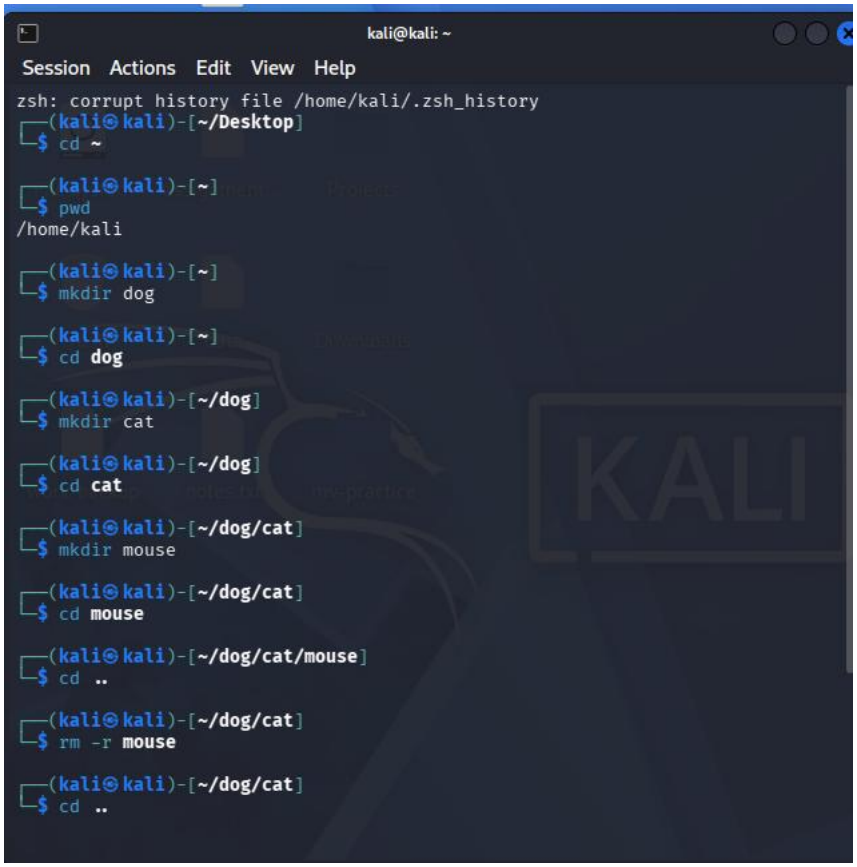
National University of Modern Language

Department of Software Engineering

Lab: # 03

Presented to:	**Ms. Rubab Mushtaq**
Presented by:	**Uzma Mumtaz**
Due Date:	**13-Oct-2026**
Roll No:	**I-SE2254426**
Subject:	**Opearting System**

Lab # 3



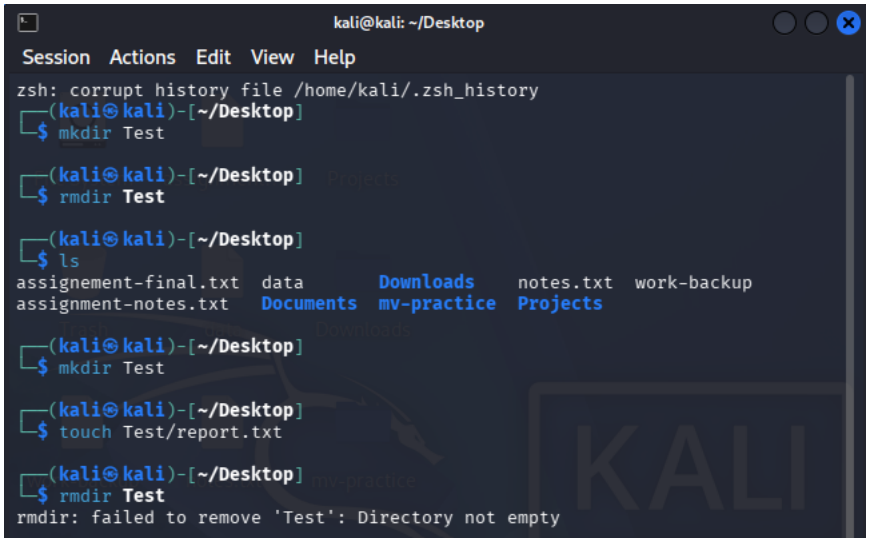


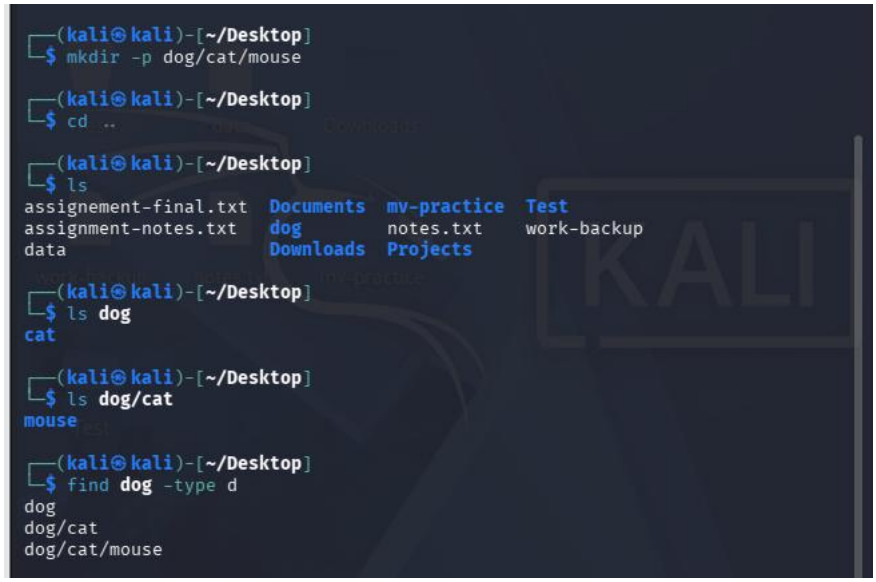
Point 3

Why rmdir is still useful when rm -r exists?

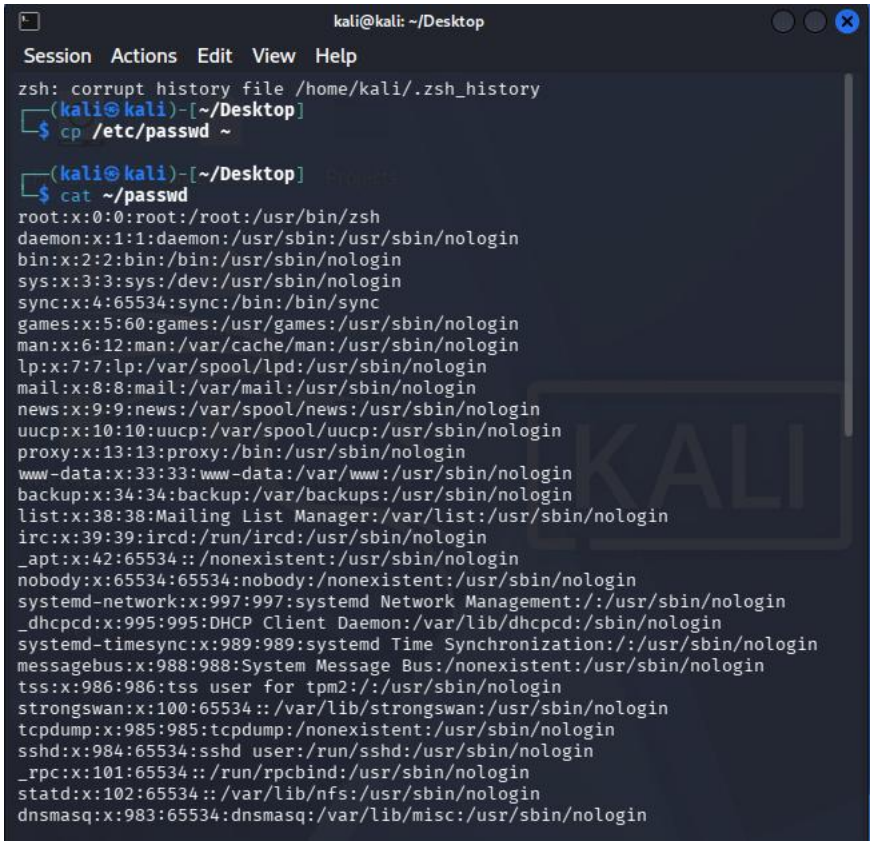
Answer

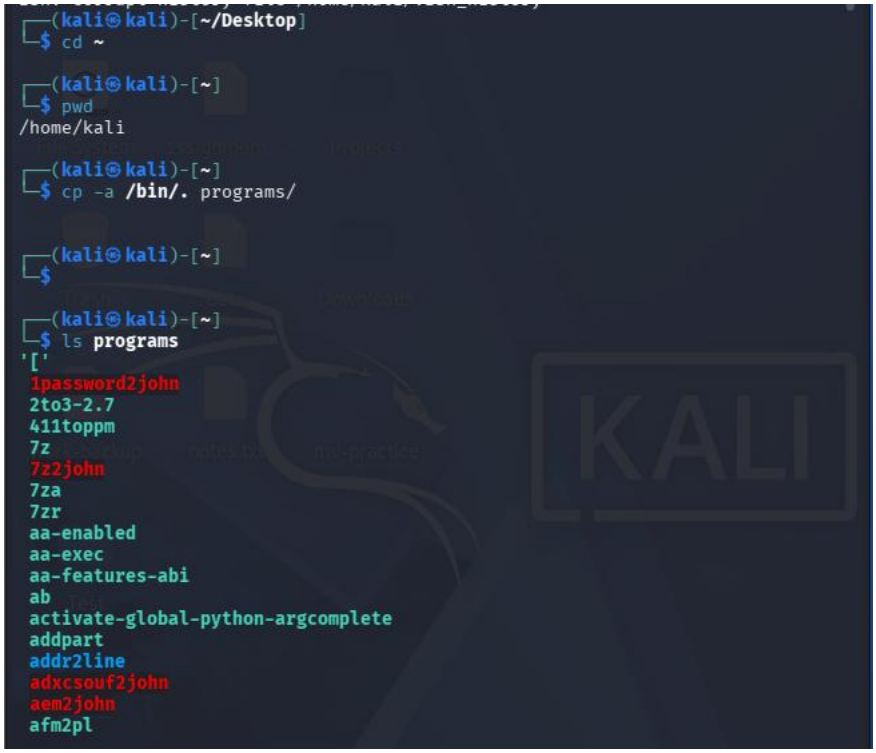
rmdir is important because it **only removes empty directories**.


kali@kali: ~/Desktop
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~/Desktop]
$ mkdir Test
(kali㉿kali)-[~/Desktop]
$ rmdir Test
(kali㉿kali)-[~/Desktop]
$ ls
assignement-final.txt data Downloads notes.txt work-backup
assignment-notes.txt Documents mv-practice Projects
(kali㉿kali)-[~/Desktop]
$ mkdir Test
(kali㉿kali)-[~/Desktop]
$ touch Test/report.txt
(kali㉿kali)-[~/Desktop]
$ rmdir Test
rmdir: failed to remove 'Test': Directory not empty


(kali㉿kali)-[~/Desktop]
$ mkdir -p dog/cat/mouse
(kali㉿kali)-[~/Desktop]
$ cd ..
(kali㉿kali)-[~/Desktop]
$ ls
assignement-final.txt Documents mv-practice Test
assignment-notes.txt dog notes.txt work-backup
data Downloads Projects
(kali㉿kali)-[~/Desktop]
$ ls dog
cat
(kali㉿kali)-[~/Desktop]
$ ls dog/cat
mouse
(kali㉿kali)-[~/Desktop]
$ find dog -type d
dog
dog/cat
dog/cat/mouse

Task # 2


kali@kali: ~/Desktop
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~/Desktop]
$ cp /etc/passwd ~
(kali㉿kali)-[~/Desktop]
$ cat ~/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:997:997:systemd Network Management:/:/usr/sbin/nologin
_dhcpcd:x:995:995:DHCP Client Daemon:/var/lib/dhcpcd:/sbin/nologin
systemd-timesync:x:989:989:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:988:988:System Message Bus:/nonexistent:/usr/sbin/nologin
tss:x:986:986:tss user for tpm2:/:/usr/sbin/nologin
strongswan:x:100:65534::/var/lib/strongswan:/usr/sbin/nologin
tcpdump:x:985:985:tcpdump:/nonexistent:/usr/sbin/nologin
sshd:x:984:65534:sshd user:/run/sshd:/usr/sbin/nologin
_rpc:x:101:65534::/run/rpcbind:/usr/sbin/nologin
statd:x:102:65534::/var/lib/nfs:/usr/sbin/nologin
dnsmasq:x:983:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin


(kali㉿kali)-[~/Desktop]
$ cd ~
(kali㉿kali)-[~]
$ pwd
/home/kali
(kali㉿kali)-[~]
$ cp -a /bin/. programs/
(kali㉿kali)-[~]
$
(kali㉿kali)-[~]
$ ls programs
'['
1password2john
2to3-2.7
411toppm
7z
7z2john
7za
7zr
aa-enabled
aa-exec
aa-features-abi
ab
activate-global-python-argcomplete
addpart
addr2line
adxcsouf2john
aem2john
afm2pl

```
┌──(kali㉿kali)-[~]
└─$ rm -rf programs/*
zsh: sure you want to delete more than 100 files in /home/kali/programs [yn]
? y

┌──(kali㉿kali)-[~]
└─$ rmdir programs

┌──(kali㉿kali)-[~]
└─$ rm users

┌──(kali㉿kali)-[~]
└─$ ls
answer.txt      file.txt    Music             report.txt
Desktop         '~fred'     Mydocuments.txt   secret-of-the-universe
docs.txt        fred        Pictures          shakespeare.txt
documents.txt   hello.txt   Projects          Templates
document.txt    Hello.txt   Public            Uzma.txt
Dog             jeff        receipe           Videos
```

Task # 3

```
┌──(kali㉿kali)-[~/Desktop]
└─$ touch baked-beans

┌──(kali㉿kali)-[~/Desktop]
└─$ ls -l baked-beans
-rw-rw-r-- 1 kali kali 0 Oct 13 12:33 baked-beans

┌──(kali㉿kali)-[~/Desktop]
└─$

┌──(kali㉿kali)-[~/Desktop]
└─$ touch -d "2026-12-31 10:00" baked-beans

┌──(kali㉿kali)-[~/Desktop]
└─$ ls -l baked-beans
-rw-rw-r-- 1 kali kali 0 Dec 31  2026 baked-beans

┌──(kali㉿kali)-[~/Desktop]
└─$ rm baked-beans

┌──(kali㉿kali)-[~/Desktop]
└─$ ls baked-beans
ls: cannot access 'baked-beans': No such file or directory

┌──(kali㉿kali)-[~/Desktop]
└─$
```